Number Sequences (Set B)

Year 4 Pt 2 · **Homework Worksheet**

Worked Example

Example — Mixed Sequence Practice: What comes next? 3, 7, 11, 15, __ and what type of sequence is this (arithmetic or geometric)?

Answer: 19 comes next (gap +4 throughout). This is an arithmetic sequence because the terms increase by a constant amount.

Quick Practice (15 questions)

1. What comes next? 5, 10, 15, 20, __
2. What comes next? 90, 80, 70, 60, __
3. What comes next? 3, 6, 12, 24, __ A) 30 B) 36 C) 42 D) 48 E) 54
4. Select ALL that apply: which of these sequences increase by exactly 3 each time? A) 1, 4, 7, 10 B) 2, 5, 9, 12 C) 0, 3, 6, 9 D) 5, 8, 10, 13 E) 10, 13, 16, 19
5. What comes next? 8, 16, 24, 32, __
6. What comes next? 81, 72, 63, 54, __ A) 44 B) 45 C) 46 D) 47 E) 48
7. What comes next? 1, 8, 27, 64, __ (these are cube numbers)
8. Fill in the missing number: 4, 8, __, 16, 20 (increasing by 4 each time)
9. What comes next? 3, 6, 12, 24, 48, __ A) 72 B) 84 C) 90 D) 96 E) 100
10. Select ALL that apply: which of these are true about the sequence 2, 6, 18, 54? A) Each term is multiplied by 3. B) Each term increases by 4. C) The next term would be 162. D) The sequence is geometric. E) The sequence is arithmetic.
11. What comes next? 4, 9, 14, 19, __
12. What comes next? 2, 2, 4, 6, 10, 16, __ (Fibonacci-style: add the two previous terms) A) 20 B) 22 C) 24 D) 26 E) 28
13. What comes next? 70, 63, 56, 49, __
14. Fill in the missing number: 3, 6, 12, __, 48 (each term doubles)
15. Select ALL that apply: which statements about the sequence 2, 4, 6, 8, 10 are true? A) These are even numbers. B) The next term is 12. C) Each term increases by a constant amount of 2. D) The gap between terms increases each time. E) The sequence is decreasing.

Mixed Practice (15 questions)

1. What comes next? 3, 5, 9, 15, 23, __ (the gap increases by 2 each time: +2, +4, +6, +8...)
2. What comes next? 162, 54, 18, 6, __
3. What comes next? 1, 4, 5, 9, 14, __ (Fibonacci-style: add the two previous terms)
4. A sequence follows the rule "double, then add 2". Starting at 2, what is the 4th term? (2, 6, 14, __)
5. Select ALL that apply: which of these sequences follow a Fibonacci-style rule? A) 2, 3, 5, 8, 13 B) 3, 6, 9, 12, 15 C) 1, 1, 2, 3, 5, 8 D) 4, 4, 8, 12, 20, 32 E) 2, 4, 8, 16, 32

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6. What comes next? 6, 9, 15, 24, 36, __ (the gap increases by 3 each time: +3, +6, +9, +12...)

7. A sequence follows the two-step rule "multiply by 3, then subtract 2". Starting at 2, what are the first four terms?

8. What comes next in this sequence of triangular numbers? 1, 3, 6, 10, 15, 21, __

9. In the sequence 2, 5, 10, 17, 26, __, what is the rule linking consecutive terms, and what comes next?

10. Select ALL that apply: which of these sequences have gaps that themselves increase by a constant amount? A) 2, 3, 5, 8, 12 (gaps 1,2,3,4) B) 5, 10, 15, 20 (gaps all 5) C) 1, 2, 4, 7, 11 (gaps 1,2,3,4) D) 3, 9, 27, 81 (gaps 6,18,54) E) 4, 7, 11, 16, 22 (gaps 3,4,5,6)

11. What comes next? 405, 135, 45, 15, __

12. A sequence starts 3, 4 and each new term is found by adding the previous two terms. What are the next three terms after 3, 4?

13. What comes next? 1, 3, 12, 60, __ (hint: each term is multiplied by an increasing amount: ×3, ×4, ×5...)

14. Select ALL that apply: which of these are geometric sequences? A) 5, 10, 20, 40 B) 4, 8, 12, 16 C) 200, 100, 50, 25 D) 3, 9, 27, 81 E) 2, 4, 6, 8

15. A sequence follows the rule "add 3, then multiply by 2". Starting at 1, find the next two terms after 1.

Standard / Advanced (15 questions)

1. What comes next in this sequence, and explain the two-step rule? 1, 4, 10, 22, 46, __

2. A sequence is defined by: term = (previous term × 2) + 1. If the first term is 3, find the first five terms of the sequence.

3. Select ALL that apply: for the sequence 2, 3, 5, 8, 12, 17, __, which of these are correct? A) The gaps are 1, 2, 3, 4, 5... B) The next term is 23. C) The sequence is geometric. D) The gap increases by 1 each time. E) The 7th term is 24.

4. A sequence is built from two interleaved patterns: odd-position terms are 2, 4, 6, 8... and even-position terms are 1, 5, 9, 13... Write out the first eight terms of the combined sequence.

5. What comes next, and what is the rule? 2, 3, 6, 10, 17, 28, __ (hint: compare to the Fibonacci idea, but check the starting numbers carefully)

6. A sequence starts at 200 and each term is found by halving the previous term and then adding 2. Find the first five terms, starting with 200.

7. Select ALL that apply: a student claims the sequence 1, 16, 81, 256, __ continues with 400. Which statements are correct? A) These are 4th powers (1^4, 2^4, 3^4, 4^4...). B) The next term should be $5^4 = 625$, not 400. C) 400 is correct because the gaps are increasing. D) The gaps (15, 65, 175) do not increase by a constant amount, ruling out a simple arithmetic pattern. E) The sequence is arithmetic.

8. A sequence follows "add the term's position number" — term 1 is 3, and each new term adds the NEXT position number. Find the first five terms starting from 3.

9. Explain why the sequence 3, 9, 18, 30, 45 is NOT a simple arithmetic or geometric sequence, then identify the actual rule and give the next term.

10. What comes next in this sequence, and explain the rule, which involves TWO alternating operations: 4, 10, 8, 14, 12, __ (hint: alternately add 6 and subtract 2)

11. A sequence is formed by adding consecutive even numbers to a running total, starting at 0: 0, +2, +4, +6, +8... giving totals 0, 2, 6, 12, 20... Explain the pattern type of these totals, and give the 6th total.

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12. Select ALL that apply: for the sequence 160, 80, 40, 20, 10, __, which of the following are true? A) Each term is half the previous term. B) The next term is 5. C) This is an arithmetic sequence. D) This is a geometric sequence. E) The common ratio is 1/2.

13. A sequence follows the rule: term(n) = term(n−1) + term(n−2) + 1, starting with 1 and 2. Find the next three terms after 1, 2.

14. What comes next, and explain the rule? 2, 3, 5, 8, 13, 21, __ (hint: this is the Fibonacci sequence starting from different numbers — check if it still applies)

15. A sequence of numbers represents the number of diagonals in polygons with 4, 5, 6, 7 sides: 2, 5, 9, 14. Identify the pattern in the GAPS and predict the number of diagonals for an 8-sided polygon.

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